

Ionic clusters as high-connectivity secondary building units for crystalline porous organic salts

Received: 16 February 2026

Accepted: 11 August 2026

Published online: 08 September 2026

Yu-Lin Lu¹, Yishuo Chen¹, Hang Qu¹, Laiyu Zhang¹,
Megan O'Shaughnessy^{1,2}, Thomas Fellowes¹✉ & Andrew I. Cooper^{1,2}✉

The solid-state assembly of crystalline porous materials is dictated by a balance between primary node–linker interactions and other weaker intermolecular forces. Metal–organic frameworks (MOFs) exploit highly directional and rigid coordination bonds that outweigh competing packing effects. This allows high-connectivity secondary building units (SBUs) to be assembled into a diverse range of architectures. By contrast, molecular crystals comprise weaker intermolecular interactions, making it challenging to realize porous packings with the highly connected reticular topologies achieved in MOFs. Here we introduce Coulombic aggregation as a means to generate primary SBU motifs. By suppressing intermolecular dispersion interactions, ammonium halide ion pairs aggregate into discrete multinuclear clusters that function as non-metal SBUs. Coupled with multitopic three-dimensional linkers, these ionic clusters create extended non-metal organic frameworks with diverse topologies, large voids and polar channels. This study extends reticular chemistry from directional chemical bonds to ‘soft’ ionic interactions, opening pathways to new porous materials and principles of solid-state assembly.

Coordinative secondary building units (SBUs) are key elements in reticular chemistry^{1,2}. They play a central role in constructing metal–organic frameworks (MOFs) with high surface areas and structural complexity, and they have also attracted interest because of their innate properties for catalysis, adsorption and framework stability³. Due to the strong directionality and fixed bonding numbers of coordination bonds, deviations from the idealized geometry impose considerable energy penalties. Frameworks constructed from coordinative SBUs are largely governed by the metal coordination environment. For example, ligands with extended π -systems often favour close packing in their molecular crystals through intermolecular dispersion interactions. However, in MOFs such as NU-1000 and PCN-224^{4,5}, the directionality and rigidity of the coordination bonds in the Zr_6 cluster prevent close packing between the ligands (Fig. 1a), thereby generating open architectures and large accessible voids. This makes MOF structures intuitively predictable from the symmetry of their components.

In reticular chemistry, the primary coordinative linking interactions and linker geometries tend to dominate; indeed, the weaker, non-linking intermolecular forces can often be ignored^{6,7}.

This is not true for molecular frameworks, such as hydrogen-bonded organic frameworks (HOFs) and crystalline porous organic salts (CPOs). These materials are defined by a delicate balance between labile primary linking interactions and other non-linking interactions^{8–12}. For molecular crystals, the absence of high-connectivity SBUs that can link multiple linkers, together with the greater impact of weaker intermolecular interactions, can often favour simple two-dimensional (2D) packing structures or interpenetrated three-dimensional (3D) structures rather than the more complex and diverse nets that are common in MOFs^{8,13}.

Recently, ammonium halide salts were shown to assemble into porous and thermodynamically stable frameworks in which ion pairs act as the primary linking motifs, unlocking applications such as

¹Materials Innovation Factory, Department of Chemistry, University of Liverpool, Liverpool, UK. ²Leverhulme Research Centre for Functional Materials Design, University of Liverpool, Liverpool, UK. ✉e-mail: thomas.j.fellowes@durham.ac.uk; aicooper@liverpool.ac.uk

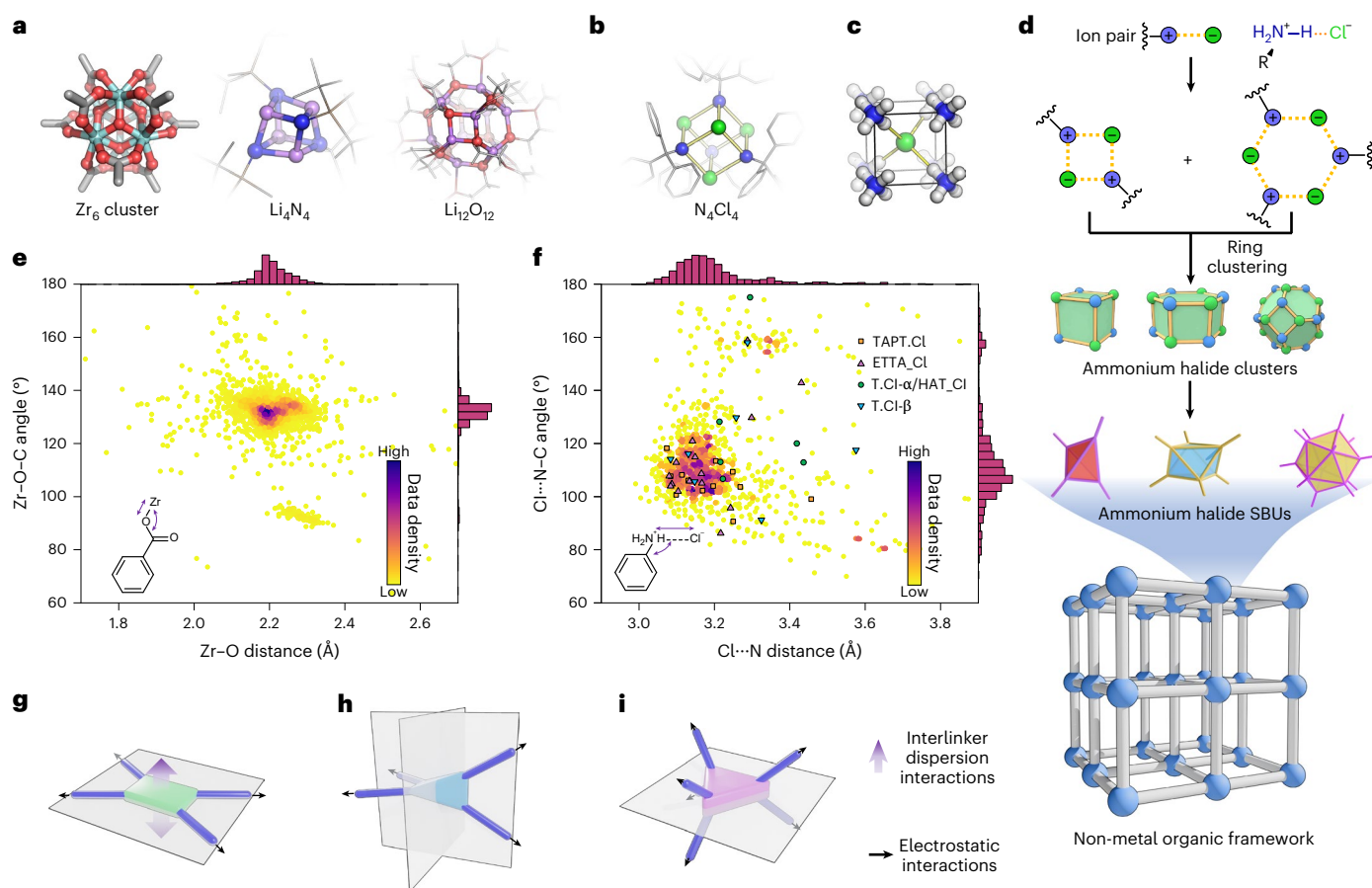


Fig. 1 | Structural design of ammonium halide SBU-based N-MOFs. a, Metal-based Zr_6 12-c SBU in MOFs compared with two representative (metal) main-group coordination clusters, cubic Li_4N_4 and twisted truncated octahedral $Li_{12}O_{12}$. **b**, Cubane-type (non-metal) ammonium halide cluster. **c**, Body-centred cubic structure of NH_4Cl ; the $Cl\cdots N$ distance is 3.34 Å. **d**, Ammonium halide SBUs, formed from aggregated ion pairs, further assemble into N-MOFs. The SBUs are defined as polyhedral assemblies, with connectivity extending from the anilinium centres. **e, f**, Distance and angle distributions of carboxylate–Zr coordination bonds (**e**) and anilinium chloride ion pairs (**f**) taken from the CSD. Previously reported porous ammonium chloride framework data are highlighted by the outlined symbols. TAPT, 1,3,5-tris(4-aminophenyl)benzene; ET TA,

4,4',4''-(ethene-1,1,2,2-tetrayl)tetraaniline; HAT and T, hexa-aminotriptycene. The marginal histograms along the top and right axes show the distributions of distances and angles, respectively. Data points are coloured according to the local point density, normalized between the calculated minimum and maximum values. **g**, Planar 4-c linker with strong interactions between linkers. **h, i**, Three-dimensional extended 4-c (**h**) and 6-c (**i**) linkers without strong interactions between linkers. In **g–i**, the charge-bearing arms of the linkers are shown as dark-blue sticks. The neutral aromatic linker cores are represented by light-green, blue, and purple polyhedra, respectively. Electrostatic and interlinker dispersion interactions are represented by black and light-purple arrows, respectively.

iodine capture, proton conduction, noble gas separation and hydrogen adsorption^{13–17}. The formation of these ion pairs is governed by Coulombic attraction, which leads to aggregation in low-dielectric media, as observed by mass spectrometry¹⁸. However, the lack of strong directionality and susceptibility to additional non-linking interactions often drives the close contacts between ion pairs into ring-laddering or ring-stacking packing modes comprising four- or six-membered rings during crystallization rather than discrete clusters¹⁹. As a result, only the simplest small N_4X_4 ionic clusters have been observed for sterically hindered ammonium halides (Fig. 1b)^{20–23}. For most ammonium halides, the packing of organic linkers leads to extended ionic $N\cdots X$ networks rather than the formation of larger, discrete N_mX_n clusters. Systematic studies by Bond revealed that the bonding geometries and electrostatic patterns of ammonium halides closely mirror those of main-group amide and imide complexes, which readily form a rich array of multinuclear clusters^{19,24}. This analogy suggests that with appropriate molecular design, ammonium halides might also be guided to assemble into discrete, high-nuclearity clustered molecular building blocks (MBBs) that could function as non-metal SBUs for the construction of 3D non-metal organic frameworks (N-MOFs) with diverse topologies (Fig. 1d).

Here we demonstrate that by controlling the geometry of organic linkers to reduce intermolecular dispersion interactions between them, we can construct symmetrical, high-connectivity ionic clusters via the aggregation of ion pairs, analogous to clusters formed through coordination bonds. The spatial separation of ionic centres enables the formation of diverse ionic clusters that can act as high-connectivity SBUs (Fig. 1d), linking the organic building blocks into a range of net topologies and thereby giving rise to 3D N-MOF structures with large voids. Energy decomposition analysis (EDA) revealed that the cluster-based assemblies show markedly reduced weak interactions between linkers compared with their non-clustered counterparts, indicating that Coulombic interactions are the dominant structural driving force. This strategy provides access to high-nuclearity ammonium halide clusters and a new perspective to expand reticular chemistry from strongly directional chemical bonds into the realm of non-directional ionic interactions.

Results and discussion

Structural design principles

Coulombic interactions between cations and anions become strong within a defined distance range, but there is relatively little dependence

on angular variation. This affords considerable structural flexibility in the interaction modes of such assemblies, making such ionic clusters sensitive to weak dispersion forces, which often dominate the packing of molecules. Previously, we addressed this by using crystal structure prediction to identify from first principles certain 2D ionic frameworks in which the thermodynamically most stable crystal packings are porous¹³, but this is expensive to generalize when one considers the large number of potential node–linker combinations and salt stoichiometries.

A Cambridge Structural Database (CSD) survey of a well-studied Zr–carboxylate metal–organic SBU showed that most Zr–O bond lengths (2.1–2.3 Å) and Zr–O–C bond angles (125–140°) are confined to a very narrow range (Fig. 1e). This reflects the directionality of metal coordination bonds, which underpins the reticular design strategy for MOFs. An equivalent analysis of aromatic primary amine $\text{Ph-NH}_3^+\cdots\text{Cl}^-$ interactions also showed favourable arrangements (Fig. 1f), but the $\text{Cl}\cdots\text{N}$ distances (3.0–3.4 Å) and $\text{Cl}\cdots\text{N-C}$ angles (80–135° and 150–160°) are distributed much more broadly than for Zr–O coordination bonds. This broader distribution highlights the challenge of constructing well-defined clusters from such flexible or ‘soft’ ionic interactions, and it explains why more complex ammonium halide clusters have remained elusive in frameworks, even though the first N_4X_4 -type ionic cluster was reported more than 60 years ago²⁰. Ingenious strategies have been devised to increase the directionality of ionic bonds using bulky groups to shield the charged centres, but this approach involves synthetic methods and so far has yielded only a limited range of connection motifs, such as linear ion pairs or 4 + 4 cluster-based linkages^{25–28}.

$\text{R-NH}_3^+\cdots\text{Cl}^-$ ion pairs tend to form multinuclear and multicentred connections with multitopic linkers. This behaviour can generate ammonium chloride-like connectivity locally (Fig. 1b,c)²⁹, which suggests that such motifs might also be used to construct more complex multinuclear ionic MBBs for framework construction^{13–15}. To realize Coulombic ionic cluster formation, it is necessary to reduce the contribution of dispersion interactions between organic linkers while also preventing multiple charged centres from connecting into extended ring-laddering ionic networks. To achieve this, the organic linkers should provide (1) multitopic connectivity, (2) rigid, 3D, spatially separated bonding directionality and (3) weak dispersion interactions between linkers. Together, these features should preserve the Coulombic attraction between charged centres and suppress the undesired stacking of linkers that might disrupt cluster formation.

Linker design and high-throughput crystallization studies

Ditopic and tritopic organic linkers typically provide linear, planar or pyramidal directionality, which can still allow close interlinker dispersion interactions in directions that are not governed by Coulombic contacts (Supplementary Fig. 15). These interactions may compete with the intended ionic connectivity and influence the resulting framework structure. By contrast, tetratopic linkers can adopt twisted tetrahedral connectivity, distributing the primary interactions in three dimensions while reducing dispersion-driven interlinker packing. We therefore considered two four-connected (4-c) linkers with similar chemical compositions but different 3D configurations. We expected the planar 5',5''-bis(4-aminophenyl)-[1,1':3'',1'':3'',1'''-quaterphenyl]-4,4'''-diamine (TAHP) linker to exhibit a strong tendency for π – π stacking (Fig. 1g and Extended Data Fig. 1a), where the final framework might be governed by linker-to-linker dispersion interactions. By contrast, the sterically hindered methyl groups in 5',5''-bis(4-aminophenyl)-2',2'',4',4'',6',6''-hexamethyl-[1,1':3'',1'':3'',1'''-quaterphenyl]-4,4'''-diamine (TAHPMe) were introduced to suppress interlinker π – π stacking, and this led to a different crystallization behaviour (Fig. 1h and Extended Data Fig. 1b). We anticipated that ammonium halide frameworks derived from TAHPMe might be less influenced by interlinker dispersion interactions and dominated by Coulombic interactions.

The strength and mode of Coulombic interactions are sensitive to the dielectric properties and proticity of the crystallization medium. This makes solvent selection important in the crystallization process. Stronger Coulombic attractions promote aggregation in low-polarity solvents with lower dielectric constants, which can therefore be used as antisolvents to tune the crystallization conditions. In this study, a dimethylformamide–methanol–antisolvent system was used for crystallization screening to balance the solubility of multiple species. The positive charge in the ammonium salt linkers originates from the protonation of the R-NH_2 centres. When multiple cationic sites are brought into close proximity, Coulombic repulsion reduces the tendency towards full protonation, rendering partially protonated species thermodynamically favourable under some conditions. We therefore systematically screened antisolvents, crystallization methods and acid equivalents, assisted by a liquid-handling robot, which allowed high-throughput mapping of the resulting polymorph landscapes (Extended Data Fig. 1).

Reaction of the planar TAHP linker with HCl generated two fully protonated structures, TAHP·Cl- α and TAHP·Cl- β , which exhibit different packing modes (Extended Data Fig. 2). TAHP·Cl- α adopts an almost planar conformation and assembles through non-porous head-to-tail packing with the linkers overall disordered (Extended Data Fig. 2a,b and Supplementary Fig. 18). When toluene was used as antisolvent under solvothermal conditions, another face-to-face packing mode was adopted, forming the TAHP·Cl- β phase with one-dimensional channels (Extended Data Fig. 2c–f). In this structure, four ammonium cations and four chloride anions form a cyclic 4 + 4 ring motif, which further packs into tubular ionic channels closely resembling the series of isorecticular structures that we reported previously¹³. Topological simplification by treating both the ionic ring and the linker as square-planar 4-c nodes gives a (4,4)-c two periodic **sql-b** net.

Introducing six methyl groups onto the two central phenyl rings in the TAHP core forces the linker to distort from square-planar connectivity into a tetrahedral geometry. The small dihedral angle present in TAHP expands to nearly 80° in TAHPMe, which is incompatible with the original cyclic ionic centre, and the ionic assembly therefore evolves into cluster-type nodes to accept connectivity from 3D space (Supplementary Figs. 16 and 17). The distortion of the linker, combined with the steric hindrance of the methyl groups, also suppresses π – π stacking of the linkers. Consequently, a diverse range of polymorphic structures was obtained, including three fully protonated forms, TAHPMe·Cl-non-cluster (nc), TAHPMe· N_4Cl_4 and TAHPMe· $\text{N}_{12}\text{Cl}_{12}$, and one partially protonated structure, TAHPMe· N_{12}Cl_7 (Fig. 2 and Supplementary Figs. 19–24).

With the exception of the non-cluster structure, TAHPMe·Cl-nc, which lacks solvent-accessible channels, the other three phases exhibit $\text{R-NH}_3^+\cdots\text{Cl}^-$ clusters constructed from the ring clustering of six- or four-membered rings and connection modes that are broadly analogous to main-group clusters such as cubic Li_4N_4 , truncated octahedral $\text{Li}_{12}\text{O}_{12}$ and hexagonal prismatic Li_6N_6 , respectively (Fig. 1a and Supplementary Figs. 23 and 27)^{30–32}. This indicates that through organic linker design, ammonium halide salts can also access larger cluster geometries reminiscent of main-group coordinative units, which has not been observed before. These highly symmetric ionic clusters can in turn function as non-metal SBUs, connecting the organic linkers into extended frameworks. Moreover, this type of 4-c linker is compatible with three distinct types of non-metal SBU, enabling the formation of three different topological structures.

Crystallization from tetrahydrofuran (THF) or 1,4-dioxane as antisolvent afforded TAHPMe· N_4Cl_4 , in which the linkers are connected by twisted cubic N_4Cl_4 clusters. These clusters can be considered as tetrahedral 4-c SBUs, giving rise to a two-fold interpenetrated (4,4)-c **dia-b** net (Fig. 2b,c and Supplementary Fig. 34). Replacing the antisolvent with dichloromethane led to a previously unreported $\text{N}_{12}\text{Cl}_{12}$ truncated octahedral cluster, which can be treated as a 12-connected

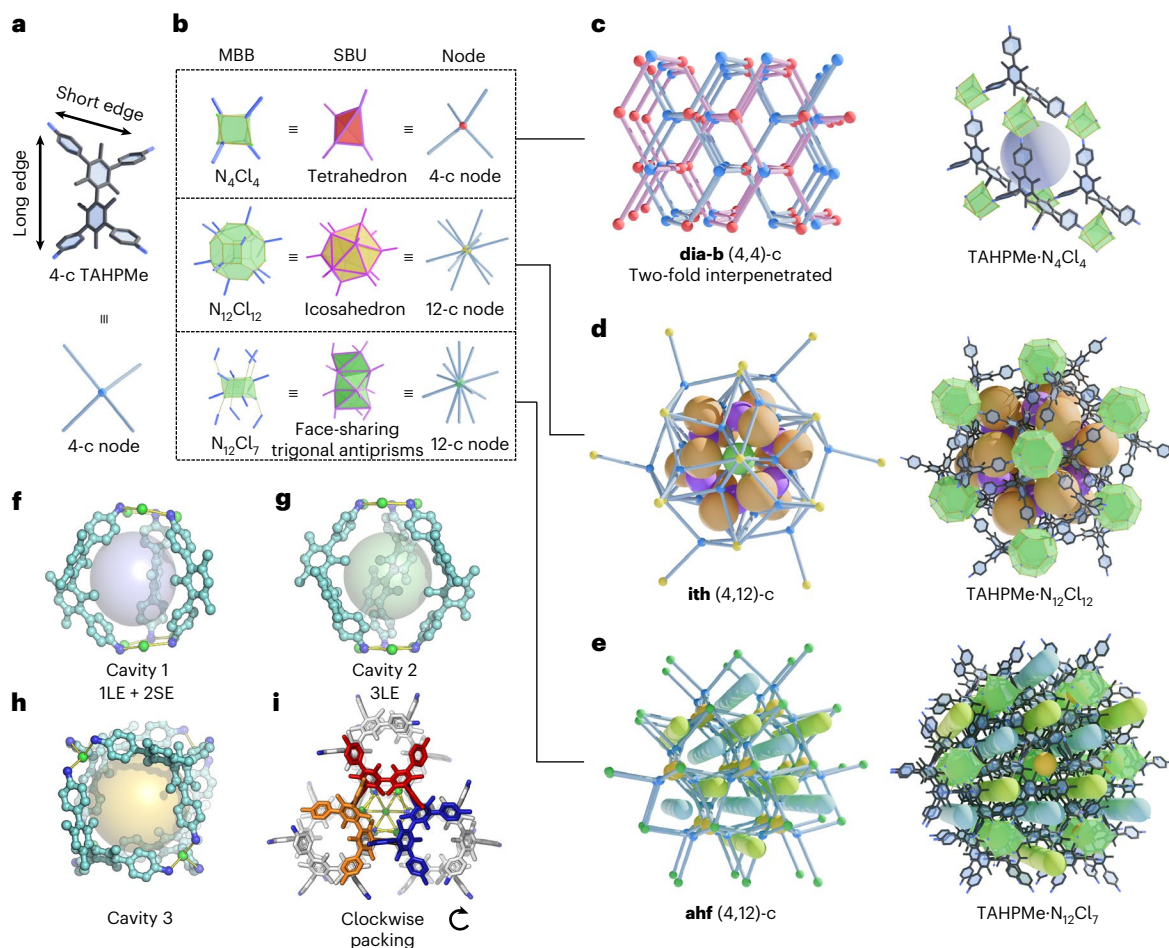


Fig. 2 | N-MOFs assembled from a 3D directional 4-c linker, TAHPMe. a, Twisted TAHPMe linker showing its long and short edges, and its representation as a 4-c node in the net. **b**, The MBBs, together with the corresponding SBUs and node connectivities. The SBU geometries were defined using the carbon atoms directly bonded to the $\text{NH}_2/\text{NH}_3^+$ groups as the points of extension. **c–e**, Topological

representations and structures of two-fold interpenetrated **dia-b** (4,4)-c TAHPMe- N_4Cl_4 (**c**), **ith** (4,12)-c TAHPMe- $\text{N}_{12}\text{Cl}_{12}$ (**d**) and **ahf** (4,12)-c TAHPMe- N_{12}Cl_7 (**e**). LE, long edge; SE, short edge. **f–h**, Cage-like structures of cavity 1 (**f**), cavity 2 (**g**) and cavity 3 (**h**) observed in TAHPMe- $\text{N}_{12}\text{Cl}_{12}$. **i**, Clockwise packing of achiral TAHPMe linkers generates the chiral centre in TAHPMe- N_{12}Cl_7 .

(12-c) icosahedral SBU, giving rise to a (4,12)-c **ith** net (Fig. 2d and Supplementary Fig. 35). This topology is difficult to realize in MOFs because the corresponding icosahedral SBUs are challenging to access^{33,34}. The short-edge and long-edge arms of the TAHPMe linkers interact with Cl^- to generate three types of cage-like cavity (Fig. 2f–h). The resulting framework resembles 3D MOFs containing cage cavities.

When crystallization was carried out in acetonitrile, ethyl acetate or diethyl ether as antisolvent, the resulting TAHPMe- N_{12}Cl_7 was not fully protonated, even with excess HCl (Fig. 2e and Supplementary Fig. 23). Six chloride anions and six protonated amines assemble into a twisted hexagonal prism with an offset ring-stacking arrangement. Within the prism, the internal cavity accommodates an additional chloride, forming an N_6Cl_7 core (Supplementary Fig. 27). Surrounding this prism core, six unprotonated amines engage with chloride ions to form a 12-c, highly symmetric N_{12}Cl_7 cluster. This twisted hexagonal prism resembles the extended ionic network recently reported for T-Cl- α (T denotes hexa-aminotriptycene)^{14,17}. However, the spatially separated linking directionality of the linkers in TAHPMe- N_{12}Cl_7 leads to the formation of discrete clusters rather than chain-like structures. This N_{12}Cl_7 cluster can be regarded as an SBU composed of three face-sharing trigonal antiprisms. As a result, the framework adopts a topology with the previously unreported point symbol of $\{4^{21}.6^{42}.8^3\}\{4^6\}_3$ in the Reticular Chemistry Structure Resource database³⁵, which we denote as the **ahf** (anilinium halide framework) net (Fig. 2e and Supplementary Fig. 36). This cluster induced spontaneous chiral clockwise or anticlockwise

packing of the achiral linker, giving rise to three-fold screw axes through the prism axis (Fig. 2i and Supplementary Figs. 22 and 23). As a result, two opposite homochiral crystals were simultaneously formed in the same vial, generating a chiral conglomerate.

Interaction analysis

EDA provides quantitative insights into the strength and nature of intermolecular weak interactions and enables their visualization via independent gradient models based on Hirshfeld partition (IGMH) analysis³⁶. We performed EDA on the constituent building blocks of the fully protonated structures to evaluate the contributions of the different intermolecular interactions (Fig. 3).

The planar TAHP exhibits the strongest pairwise total interaction energy of -117 kJ mol^{-1} between adjacent linkers, and IGMH analysis shows that this dispersion energy is mainly generated from face-to-face π - π stacking, confirming that strong dispersion interactions between the organic linkers dominate its packing mode (Fig. 3a). By contrast, for TAHPMe-Cl-nc, the total interaction energy between the non-linking fragments is -75 kJ mol^{-1} , and IGMH analysis indicates that the interactions between linkers are much weaker than those in the planar TAHP linker. These weak interactions originate mostly from CH- π contacts rather than from the π - π stacking observed in TAHP because the steric hindrance generated by the methyl groups prevents close contacts between the aromatic rings. (Fig. 3b and Supplementary Fig. 38). The framework structures TAHPMe- N_4Cl_4 and TAHPMe- $\text{N}_{12}\text{Cl}_{12}$ display

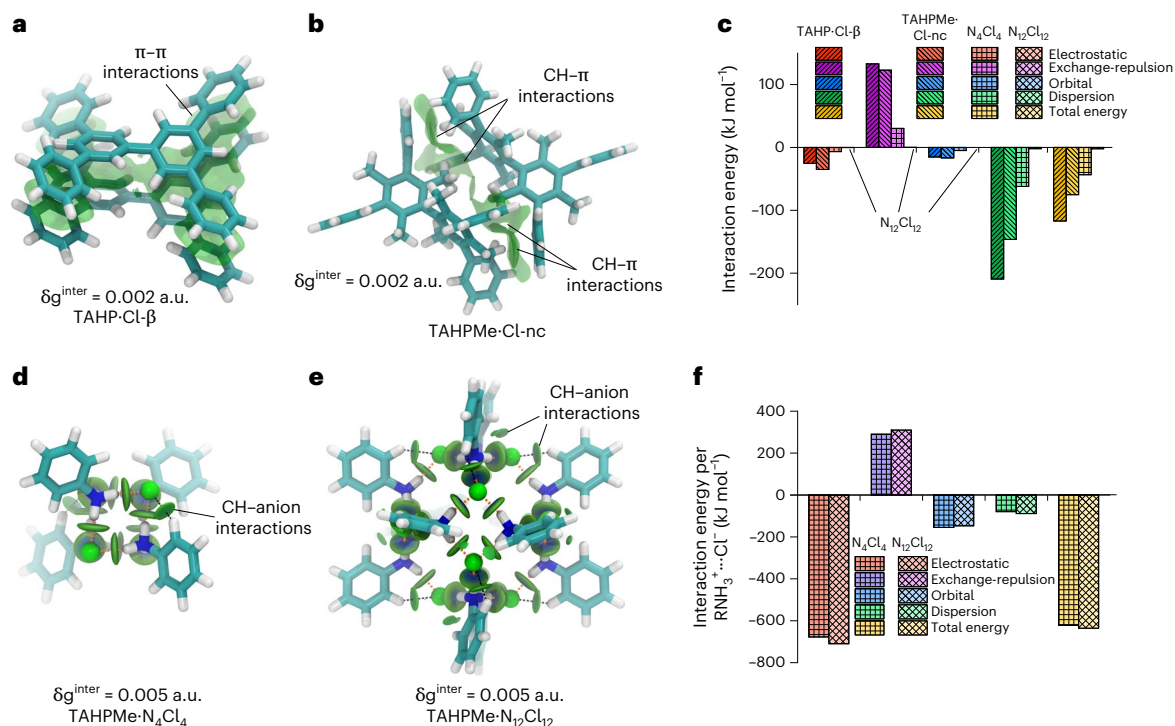


Fig. 3 | Interaction energy analyses. **a, b**, IGMH isosurfaces ($\delta g^{\text{inter}} = 0.002$ a.u.; green surfaces; δg^{inter} denotes the IGMH function used to reveal interfragment interaction regions) of the organic linkers in TAHP-Cl- β (**a**) and TAHPMe-Cl-nc (**b**). **c**, EDA of two linker fragments in TAHP-Cl- β , TAHPMe-Cl-nc, TAHPMe- N_4Cl_4 and

TAHPMe- $\text{N}_{12}\text{Cl}_{12}$. **d, e**, IGMH isosurfaces ($\delta g^{\text{inter}} = 0.005$ a.u.) of the isolated ionic clusters in TAHPMe- N_4Cl_4 (**d**) and TAHPMe- $\text{N}_{12}\text{Cl}_{12}$ (**e**). **f**, EDA of the average energy per ion pair within the ionic clusters of TAHPMe- N_4Cl_4 and TAHPMe- $\text{N}_{12}\text{Cl}_{12}$.

weaker linker–linker interactions. In particular, TAHPMe- $\text{N}_{12}\text{Cl}_{12}$ shows negligible interactions between linkers due to the absence of close contacts between linkers in the framework (Supplementary Fig. 39).

To further resolve the intermolecular interactions associated with the linkers, we also carried out EDA on two isolated fully protonated ionic clusters, N_4Cl_4 and $\text{N}_{12}\text{Cl}_{12}$ (Fig. 3d,e). Each ammonium unit and each chloride ion was treated as an individual fragment; hence, the N_4Cl_4 and $\text{N}_{12}\text{Cl}_{12}$ clusters were described by 8 and 24 fragments, respectively. Regarding the average energy of each $\text{R-NH}_3^+\cdots\text{Cl}^-$ ion pair, these clusters are overwhelmingly stabilized, as might be expected, by electrostatic attraction, with other contributions constituting only a minor part of the overall interaction energy.

Expansion to a six-connected 3D extended linker

Next, we extended the approach used for TAHPMe to higher-connectivity, elongated triptycene-based six-connected (6-c) 4,4',4'',4''',4''',4''''-(9,10-dihydro-9,10-[1,2]benzenoanthracene-2,3,6,7,14,15-hexayl)hexaaniline (TPA) systems, in which steric congestion between the elongated *ortho*-phenyl substituents enforces a non-planar geometry that suppresses undesirable π – π stacking of the linkers (Figs. 1i and 4a). When we again attempted high-throughput crystallization experiments, the low solubility of TPA precluded the formation of large crystals under solvothermal conditions. However, vapour diffusion crystallization yielded three distinct structures, each exhibiting different protonation states. At low HCl equivalents, unprotonated crystals were obtained using diethyl ether and THF as the antisolvent (Supplementary Fig. 25). The crystallization of pure TPA under identical conditions also led to the same phase, confirming that no protonation had occurred. Even without Coulombic interactions to direct the crystallization, there were no notable π – π interactions between the triptycene cores of TPA in this unprotonated structure. The main driving forces for crystallization arise instead from hydrogen bonding between amines and interlocking between neighbouring phenyl rings.

Protonation occurred using an excess of HCl and chloroform as the antisolvent, and Coulombic interactions directed the TPA molecules into a highly symmetric packing mode in the $P6_3/mmc$ space group, yielding hexagonal prismatic crystals (Fig. 4b). This framework is constructed from 6-c TPA linkers and 6-c N_6Cl_7 clusters with trigonal antiprismatic connectivity. The simplified topology can therefore be assigned to a (6,6)-c **nia** net (Fig. 4a,d,e and Supplementary Fig. 37).

We note that TPA- N_6Cl_7 adopts an ionic centre similar to that of TAHPMe- $\text{N}_{12}\text{Cl}_{12}$, in which six ammonium termini interact with six chloride anions to form a twisted hexagonal prism (Supplementary Fig. 27). However, in 6,6-c TPA- N_6Cl_7 , no additional unprotonated amines surround the prism; instead, the periphery is filled with highly disordered solvent molecules. This lack of auxiliary linker interactions renders the resulting crystals susceptible to changes in the solvent environment. Even gentle washing with the crystallization antisolvent led to fragmentation of the hexagonal crystals.

The linking pattern became more complex when an insufficient stoichiometry of HCl was used in chloroform. The lack of chloride anions prevents the $\text{R-NH}_3^+\cdots\text{Cl}^-$ and $\text{R-NH}_2\cdots\text{Cl}^-$ combinations from assembling into a highly symmetric cluster in this system. As a result, an irregular polygonal N_{12}Cl_9 cluster caused the TPA salt to adopt a lower-symmetry monoclinic $P2_1$ space group, giving rise to rod-shaped crystals (Fig. 4c,f,g and Supplementary Figs. 28–30). This TPA- N_{12}Cl_9 phase can also be obtained by recrystallizing the TPA- N_6Cl_7 sample after heating it under vacuum for 24 h. Owing to the irregularity of the cluster geometry and the broad distribution of $\text{Cl}\cdots\text{N}$ distances, small changes in the distance threshold can lead to markedly different topological models (Supplementary Figs. 32 and 33). We therefore did not assign a topology to this structure.

Owing to the lack of interlinker constraints, the phenyl rings in the highly symmetric TPA- N_6Cl_7 structure possess two-fold rotational disorder (Supplementary Fig. 31). By contrast, in the partially protonated TPA- N_{12}Cl_9 , the higher amine/chloride ratio provides extra

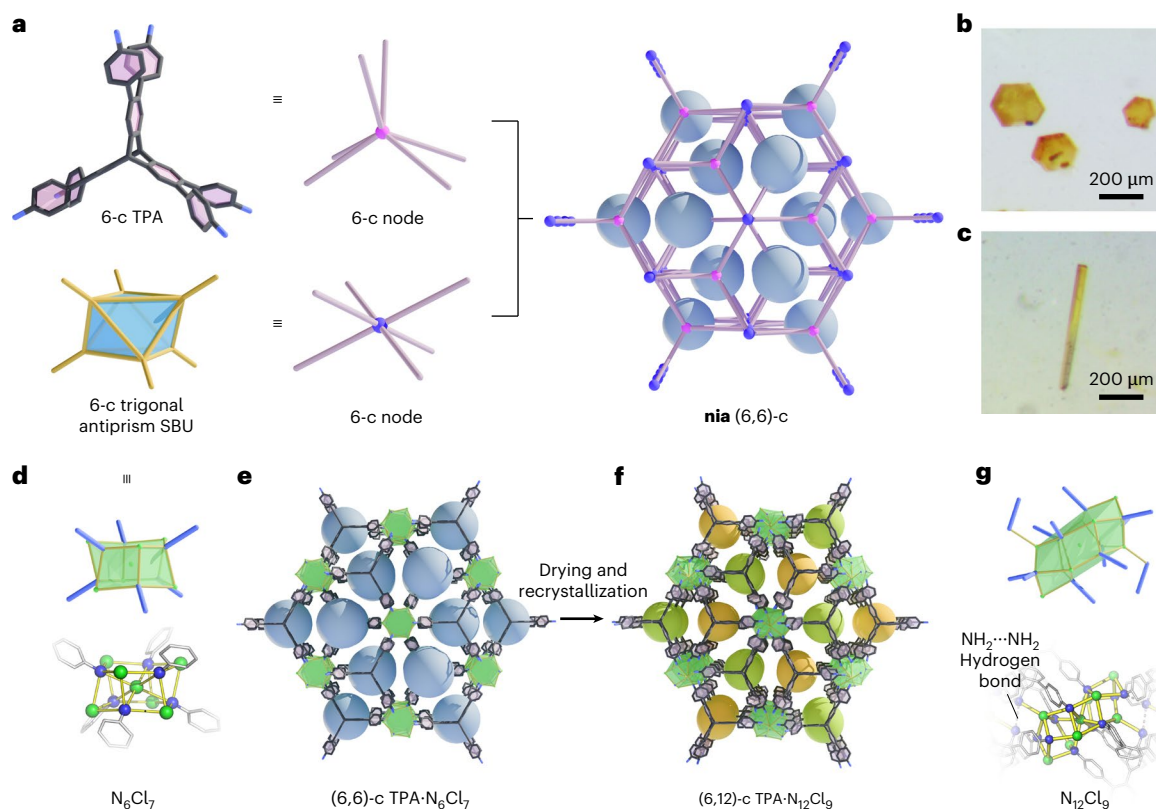


Fig. 4 | N-MOFs assembled from 3D, spatially directional 6-c TPA. **a**, Structures of the TPA linker and the 6-c N_6Cl_7 SBU, with their corresponding node connectivity. These two 6-c nodes can form a (6,6)-c *nia* net. **b,c**, Images of crystals of $TPA \cdot N_6Cl_7$ (**b**) and $TPA \cdot N_{12}Cl_9$ (**c**). **d**, Twisted hexagonal prism structure

of the N_6Cl_7 cluster. **e,f**, Crystal structures of $TPA \cdot N_6Cl_7$ (**e**) and $TPA \cdot N_{12}Cl_9$ (**f**). **g**, Irregular geometry of the $N_{12}Cl_9$ cluster. The yellow sticks indicate electrostatic and hydrogen-bonding interactions between N and Cl centres ($N \cdots Cl$ distances $< 3.4 \text{ \AA}$).

amine binding sites for each chloride; together with the interlocking of neighbouring linkers, this effectively fixes the molecular conformation and suppresses rotational disorder. As a result, $TPA \cdot N_{12}Cl_9$ shows higher stability than $TPA \cdot N_6Cl_7$, and retains good crystallinity even after drying under vacuum for 10 h (Supplementary Fig. 44).

Crystal stability and guest-binding properties

Compared with most previously reported ionic molecular frameworks, the above structures exhibit remarkably large and well-defined solvent accessible voids. However, porous salts are highly sensitive to solvent loss, which often leads to framework collapse⁹. We examined the stability of these materials after vacuum drying and after supercritical CO_2 (sc CO_2) drying. Among the samples tested, only $TAHPMe \cdot N_{12}Cl_7$ and $TPA \cdot N_{12}Cl_9$ retained crystallinity after solvent removal; for the other materials, diffraction peaks were lost upon drying (Supplementary Figs. 40–45). This shows that, even though these frameworks are constructed through multiple strong ionic linkages, a proportion of them remain highly susceptible to framework collapse upon solvent removal.

$TAHPMe \cdot N_{12}Cl_{12}$ was observed to lose its crystallinity after direct removal of most exchanged solvents (including dichloromethane, diethyl ether, ethyl acetate or THF), while this cavity-containing framework exhibited better stability after solvent exchange with *n*-pentane or cyclohexane (Extended Data Fig. 3a). Even after 12 h under dynamic vacuum at room temperature, the *n*-pentane-loaded sample retained its crystallinity. Moreover, NMR analysis showed that about 14 wt% of *n*-pentane remained in the framework, indicating that the cavities effectively trap *n*-pentane molecules within the structure (Supplementary Fig. 46). Only when the *n*-pentane@ $TAHPMe \cdot N_{12}Cl_{12}$ sample was heated at 393 K for 4 h did the diffraction

signals gradually diminish (Extended Data Fig. 3b). This demonstrates that these low-boiling hydrocarbons (*n*-pentane, boiling point 309 K) can stabilize the $TAHPMe \cdot N_{12}Cl_{12}$ framework through host–guest interactions. However, when we tried to remove the *n*-pentane using sc CO_2 , the framework still collapsed (Supplementary Fig. 47), indicating that even the use of a zero-surface-tension extraction medium could not preserve the latent porosity in $TAHPMe \cdot N_{12}Cl_{12}$. Kinetic thermogravimetric analysis measurements at 333 K further revealed that a partial release of *n*-pentane occurs during heating, which correlates well with the decrease in the *n*-pentane ratio in the structure as determined by NMR analysis (Supplementary Fig. 48), confirming that the encapsulated hydrocarbons exhibit slow release kinetics from the cage cavities.

To investigate the guest-binding modes within the framework, we exchanged the crystal samples with *n*-pentane, *n*-heptane, cyclohexane, methylcyclohexane and carvone. However, apart from the *n*-heptane- and cyclohexane-loaded samples, the guest molecules could not be resolved unambiguously because of their disorder within the highly symmetric framework. We also noted that the crystallinity of $TAHPMe \cdot N_{12}Cl_{12}$ improved when guests were included, yielding crystals with enhanced diffraction quality (Supplementary Fig. 49). This observation suggests that the inclusion of guest molecules promotes structural ordering in the framework. For the *n*-heptane-loaded structure, the linear alkane molecules interact through multiple CH– π and van der Waals contacts between the terminal methyl groups and the two N_3Cl_3 rings of cavity 1, demonstrating a well-defined and size-matched host–guest interaction pattern (Extended Data Fig. 3c). Cyclohexane inclusion presented a more intriguing case as two cyclohexane molecules can simultaneously occupy one cavity 1, forming distinct CH– π contacts with surrounding aromatic rings (Extended Data Fig. 3d).

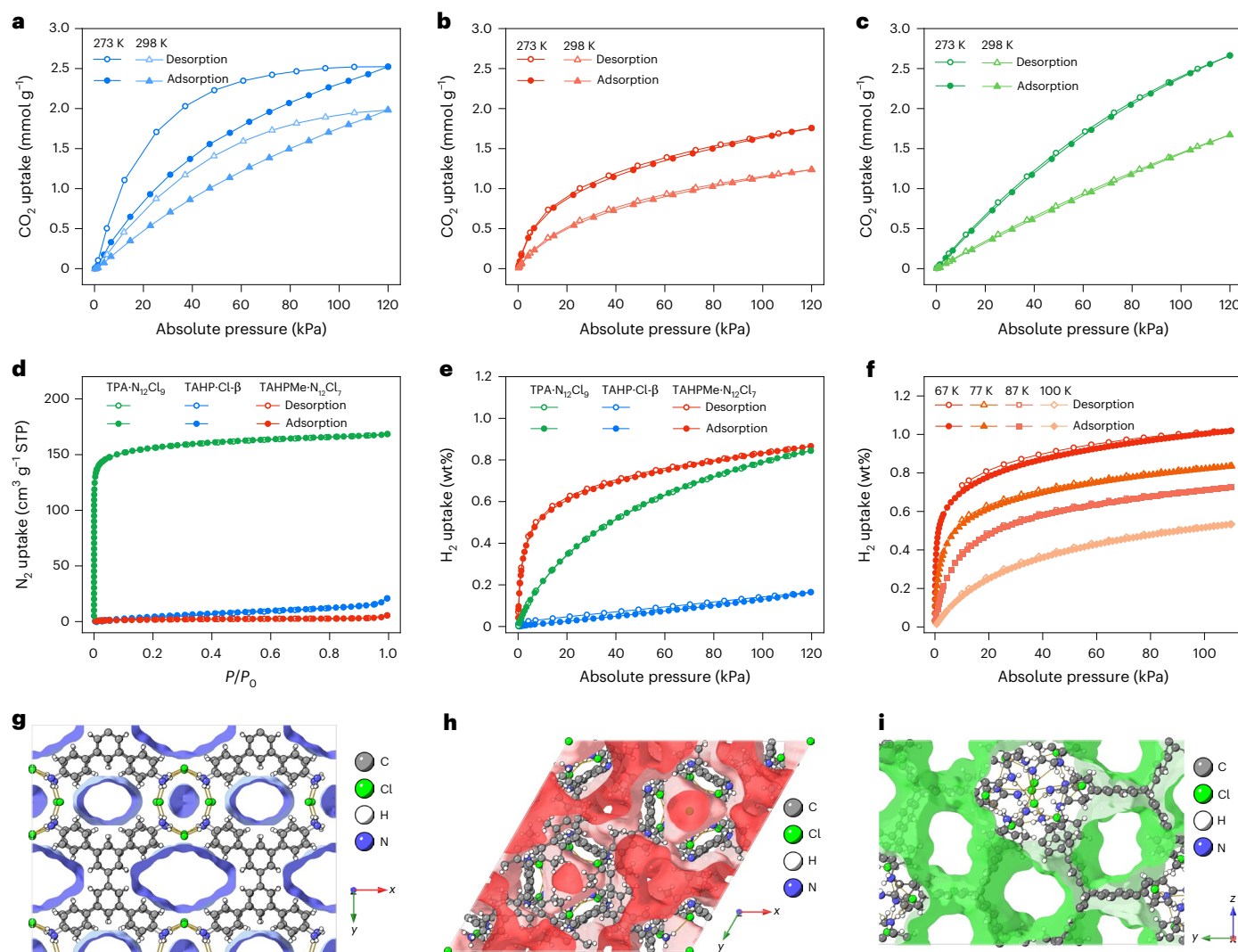


Fig. 5 | Gas adsorption and pore characteristics of the stable N-MOFs. a–c, CO_2 adsorption isotherms of TAHP-Cl- β (**a**), TAHPMe- Ni_2Cl_7 (**b**) and TPA- Ni_2Cl_9 (**c**). **d**, N_2 adsorption isotherms of TAHP-Cl- β , TAHPMe- Ni_2Cl_7 and TPA- Ni_2Cl_9 at 77 K as a function of relative pressure (P/P_0). STP, standard temperature and pressure; P_0 ,

saturation vapour pressure of the adsorbate at the measurement temperature. **e**, H_2 adsorption isotherms of TAHP-Cl- β , TAHPMe- Ni_2Cl_7 and TPA- Ni_2Cl_9 at 77 K. **f**, H_2 adsorption isotherms of TAHPMe- Ni_2Cl_7 at 67, 77, 87 and 100 K. **g–i**, Accessible voids in TAHP-Cl- β (**g**), TAHPMe- Ni_2Cl_7 (**h**) and TPA- Ni_2Cl_9 (**i**).

Gas adsorption properties

The porosity characteristics of the structurally stable frameworks were further investigated by gas adsorption experiments (Fig. 5). CO_2 adsorption was first examined at 273 and 298 K. TAHP-Cl- β , TAHPMe- Ni_2Cl_7 and TPA- Ni_2Cl_9 exhibited CO_2 uptakes of 2.3, 1.6 and 2.4 mmol g^{-1} , respectively, at 273 K and 100 kPa (Fig. 5a–c). TAHP-Cl- β exhibited a pronounced desorption hysteresis, which we attribute to the incomplete removal of strongly bound toluene guests residing in the channels (Supplementary Fig. 12). As a consequence, TAHP-Cl- β displays relatively slow adsorption kinetics. This reduced mobility was more pronounced at 195 K, resulting in a lower adsorption capacity compared with that observed at 273 K (Supplementary Fig. 50). For N_2 adsorption at 77 K, TAHPMe- Ni_2Cl_7 and TAHP-Cl- β exhibited negligible N_2 uptake, while TPA- Ni_2Cl_9 showed a type-I isotherm, reaching 168.4 $\text{cm}^3 \text{g}^{-1}$ (standard temperature and pressure) at saturation (Fig. 5d). Unlike MOFs, nitrogen adsorption at 77 K is relatively uncommon in polar porous organic salts⁹. The adsorption of N_2 in TPA- Ni_2Cl_9 may originate from its comparatively large pore dimensions, together with the presence of interlocked phenyl rings that embed the charged sites within the framework, thereby reducing the polarity of the accessible voids. The substantial uptake at low pressure confirms the ultra-microporosity

of this material, and the fitted pore-size distribution reveals a pore width of 6.5 Å (Supplementary Fig. 52 and Supplementary Table 7). Brunauer–Emmett–Teller analysis gave a surface area of 611 $\text{m}^2 \text{g}^{-1}$ (Supplementary Fig. 53).

For hydrogen adsorption at 77 K (Fig. 5e), TAHPMe- Ni_2Cl_7 exhibited an uptake of 0.83 wt%, surpassing that of TPA- Ni_2Cl_9 , despite the latter having both a high surface area and large cavities. Fitting the H_2 isotherm gave a pore size of about 3.8 Å, closely matching the pore-limiting diameter determined from its crystal structure (Supplementary Fig. 54). The pronounced uptake at low pressures suggests strong interactions of H_2 within the narrow polar channels. Adsorption isotherms collected at 67, 77, 87 and 100 K gave an isosteric heat of adsorption of 8.0 kJ mol^{-1} for H_2 at low loadings, which exceeds that of most MOFs, HOFs and covalent organic frameworks (COFs) (Fig. 5f, Supplementary Fig. 55 and Supplementary Table 8)³⁷. This high affinity for H_2 likely originates from both van der Waals and electrostatic interactions between the confined H_2 and the charged narrow channel, where quadrupole charge coupling and charge-induced dipole interactions can increase the binding affinity, which decays rapidly with distance between H_2 and the charged centres³⁸.

Conclusions and outlook

We have demonstrated herein a series of N-MOF structures constructed from non-metal SBUs that are distinct from conventional HOFs, CPOSSs and other porous molecular crystals. These frameworks are assembled from multinuclear, high-connectivity nodes composed of Coulombically linked ion pairs, endowing them in some cases with large voids, 3D channels and complex nets, analogous to those of classical 3D MOFs. The large cage-like cavity in TAHPMe-N₁₂Cl₁₂ enables the binding of alkanes, whereas the narrow and highly polar channels in TAHPMe-N₁₂Cl₇ impart a pronounced affinity for hydrogen.

EDA revealed that minimizing the dispersion-dominated non-linking interactions between linkers enables electrostatic attraction of the ammonium chloride ion pairs to dominate, directing cluster formation. The resulting ionic clusters act as non-metal SBUs, connecting three-dimensionally oriented multitopic cationic linkers into extended networks.

The non-metal SBUs adopt connection motifs that are reminiscent of main-group coordination clusters and which are difficult to realize in conventional ammonium halide systems. With the exception of one low-symmetry N₁₂Cl₉ cluster, all the cluster structures follow a clear geometric pattern, each being constructed from quadrilateral and hexagonal faces. Among regular and semi-regular polyhedra, only the cubic, hexagonal prismatic and truncated octahedral forms, which are those obtained in this work, can be constructed solely from quadrilateral and hexagonal faces. This geometrical correspondence indicates that employing ammonium halides as building blocks may afford a certain level of predictability in the assembly of these cluster architectures, notwithstanding that CSD suggest that these clusters are much ‘softer’ SBUs than those found in MOFs (Fig. 1e,f). This finding demonstrates that the design logic for reticular chemistry can be extended beyond coordination bonding to an electrostatically governed ionic cluster regime, at least to some extent. However, framework formation in N-MOFs derived from non-metal SBUs is sensitive to the influence of weak interactions and crystallization solvent, which results in the broad diversity of structural architectures found here, even though we explored only a relatively small area of chemical space.

Our results indicate that the formation of the ‘softer’ ionic cluster motifs reported here is governed primarily by the structural features of the linker. Thus, once a given linker type is shown to support ionic cluster formation as an SBU, it may provide a platform for accessing other cluster architectures with different anions. For example, other studies have demonstrated that other anions, such as HSO₄[−] and NO₃[−], also tend to aggregate with multitopic cations³⁹. This situation parallels the early development stages of first- and second-generation MOFs, suggesting that the electrostatically governed N-MOFs could hold potential for future advances, including exploration of the specific properties of the constituent ionic clusters, for example, in adsorption and separation processes, or to introduce specific framework dynamics. In addition, different possible stoichiometries in ionic clusters enable the incorporation of auxiliary linkers. For instance, although the N₁₂Cl₇ and N₆Cl₇ clusters share an identical hexagonal prismatic core, the presence of auxiliary linkers in N₁₂Cl₇ confers greater structural stability. This feature also suggests that established strategies for the structural modification of MOFs, such as spacer installation⁴⁰ or pore space partition⁴¹, might be translated to these ionic clusters in the future.

The facile crystallization of these materials under mild conditions makes them well suited for high-throughput approaches, allowing us to map and expand this chemical space. Such strategies will be valuable both for developing crystalline porous materials with diverse pore architectures and for gaining a deeper fundamental understanding of these Coulomb-directed materials.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-026-02248-w>.

References

1. Yaghi, O. M., Kalmutzki, M. J. & Diercks, C. S. *Introduction to Reticular Chemistry: Metal-Organic Frameworks and Covalent Organic Frameworks* (Wiley, 2019).
2. Jiang, H. et al. Merged-nets enumeration for the systematic design of multicomponent reticular structures. *Science* **386**, 659–666 (2024).
3. Kalmutzki, M. J., Hanikel, N. & Yaghi, O. M. Secondary building units as the turning point in the development of the reticular chemistry of MOFs. *Sci. Adv.* **4**, eaat9180 (2018).
4. Feng, D. et al. Construction of ultrastable porphyrin Zr metal–organic frameworks through linker elimination. *J. Am. Chem. Soc.* **135**, 17105–17110 (2013).
5. Mondloch, J. E. et al. Vapor-phase metalation by atomic layer deposition in a metal–organic framework. *J. Am. Chem. Soc.* **135**, 10294–10297 (2013).
6. Jiang, H., Alezi, D. & Eddaoudi, M. A reticular chemistry guide for the design of periodic solids. *Nat. Rev. Mater.* **6**, 466–487 (2021).
7. Guillermin, V., Jiang, H., Alezi, D., Alsadun, N. & Eddaoudi, M. From elementary to advanced design of functional metal–organic frameworks: a user guide to deciphering the reticular chemistry toolbox. *Adv. Mater.* **37**, e2414153 (2025).
8. Lin, R.-B. & Chen, B. Hydrogen-bonded organic frameworks: chemistry and functions. *Chem* **8**, 2114–2135 (2022).
9. Xing, G., Peng, D. & Ben, T. Crystalline porous organic salts. *Chem. Soc. Rev.* **53**, 1495–1513 (2024).
10. Li, L. et al. Flue gas desulfurization and SO₂ recovery within a flexible hydrogen-bonded organic framework. *Nat. Chem.* **17**, 727–733 (2025).
11. White, N. G. & McGuirk, C. M. Defining, designing and determining the structure of supramolecular frameworks. *Chem. Soc. Rev.* **54**, 9612–9629 (2025).
12. Li, P., Ryder, M. R. & Stoddart, J. F. Hydrogen-bonded organic frameworks: a rising class of porous molecular materials. *Acc. Mater. Res.* **1**, 77–87 (2020).
13. O’Shaughnessy, M. et al. Porous isorecticular non-metal organic frameworks. *Nature* **630**, 102–108 (2024).
14. O’Shaughnessy, M. et al. Polar triptycene-based nonmetal organic frameworks show enhanced hydrogen adsorption. *J. Am. Chem. Soc.* **147**, 39351–39358 (2025).
15. O’Shaughnessy, M. et al. Nonmetal organic frameworks exhibit high proton conductivity. *J. Am. Chem. Soc.* **147**, 15429–15434 (2025).
16. Xie, Y., Ding, X., Wang, J. & Ye, G. Hydrogen-bonding assembly meets anion coordination chemistry: framework shaping and polarity tuning for xenon/krypton separation. *Angew. Chem. Int. Ed.* **62**, e202313951 (2023).
17. Xie, Y., Gao, Q., Wang, J., Ye, G. & Chen, B. Guest polarizability directed molecular ‘freezing’ within non-metal porous salt frameworks. *Angew. Chem. Int. Ed.* **64**, e202509905 (2025).
18. Lentz, N. B. & Houk, R. S. Negative ion mode electrospray ionization mass spectrometry study of ammonium-counter ion clusters. *J. Am. Soc. Mass Spectrom.* **18**, 285–293 (2007).
19. Bond, A. D. Organic ammonium halides as analogues of main-group amide/imide complexes in the solid state: extension of ring-laddering and ring-stacking principles. *Coord. Chem. Rev.* **249**, 2035–2055 (2005).
20. Dobler, M. & Dunitz, J. D. Die konformation des bicyclo [3.3.1] nonan-systems. I. strukturanalyse des 3-azabicyclo [3.3.1] nonan-hydrobromids. *Helv. Chim. Acta* **47**, 695–704 (1964).

21. Bond, A. D. & Doyle, E. L. Hydrogen-bonded cubanes in the crystal structure of 2,6-di(Pr)aniline hydrochloride and their inorganic analogues $[M^{2+}(2,6\text{-di(Pr)C}_6\text{H}_5\text{N}^2\text{)}]_4$ (M = Sn, Pb). *Chem. Commun.* **39**, 2324–2325 (2003).
22. Bond, A. D., Jorgensen, W. H. & Ploger, J. M. Hydrogen-bonded cubanes and ladder fragments by analogy with the inorganic solid state. *Chem. Commun.* **43**, 3273–3275 (2007).
23. Huber, T. et al. Where does the proton go? Structure and dynamics of hydrogen-bond switching in aminophosphine chalcogenides. *Angew. Chem. Int. Ed.* **64**, e202425049 (2025).
24. Bond, A. D. Structural motifs in secondary ammonium halides: ring-stacking and ring-laddering in the organic solid state. *Chem. Eur. J.* **10**, 1885–1898 (2004).
25. Ami, T., Oka, K., Tsuchiya, K. & Tohnai, N. Porous organic salts: diversifying void structures and environments. *Angew. Chem. Int. Ed.* **61**, e202202597 (2022).
26. Hutskalov, I., Linden, A. & Coric, I. Directional ionic bonds. *J. Am. Chem. Soc.* **145**, 8291–8298 (2023).
27. Yamamoto, A. et al. Diamondoid porous organic salts toward applicable strategy for construction of versatile porous structures. *Cryst. Growth Des.* **12**, 4600–4606 (2012).
28. Miyano, T. et al. A structurally variable porous organic salt based on a multidirectional supramolecular cluster. *Chem. Eur. J.* **22**, 15430–15436 (2016).
29. Vainshtein, B. Refinement of the structure of the group NH_4 in the structure of ammonium chloride. *Tr. Inst. Kristallogr. Akad. Nauk SSSR* **12**, 18–24 (1956).
30. Barr, D., Clegg, W., Mulvey, R. E., Snaith, R. & Wade, K. Bonding implications of interatomic distances and ligand orientations in the iminolithium hexamers $[\text{LiN}=\text{C}(\text{Ph})\text{Bu}]_6$ and $[\text{LiN}=\text{C}(\text{Ph})\text{NMe}_2]_6$: a stacked-ring approach to these and related oligomeric organolithium systems. *J. Chem. Soc. Chem. Commun.* **22**, 295–297 (1986).
31. Ruwisch, L., Klingebiel, U., Rudolph, S., Herbst-Irmer, R. & Noltemeyer, M. Di-tert-butylmethylsilylamide—a building block for azasilacyclobutane synthesis. *Chem. Ber.* **129**, 823–828 (1996).
32. Randazzo, J., Jacob Morris, J., Rood, J. A., Noll, B. C. & Henderson, K. W. Lithium-mediated ring opening of 1,4-dioxane and structural characterization of a $\text{Li}_{12}\text{O}_{12}$ truncated octahedron. *Inorg. Chem. Commun.* **11**, 1270–1272 (2008).
33. Chen, Z. et al. Reticular chemistry in the rational synthesis of functional zirconium cluster-based MOFs. *Coord. Chem. Rev.* **386**, 32–49 (2019).
34. Furukawa, H. et al. Water adsorption in porous metal–organic frameworks and related materials. *J. Am. Chem. Soc.* **136**, 4369–4381 (2014).
35. O’Keeffe, M., Peskov, M. A., Ramsden, S. J. & Yaghi, O. M. The Reticular Chemistry Structure Resource (RCSR) database of, and symbols for, crystal nets. *Acc. Chem. Res.* **41**, 1782–1789 (2008).
36. Lu, T. & Chen, Q. Independent gradient model based on Hirshfeld partition: a new method for visual study of interactions in chemical systems. *J. Comput. Chem.* **43**, 539–555 (2022).
37. Zhao, D., Wang, X., Yue, L., He, Y. & Chen, B. Porous metal–organic frameworks for hydrogen storage. *Chem. Commun.* **58**, 11059–11078 (2022).
38. Lochan, R. C. & Head-Gordon, M. Computational studies of molecular hydrogen binding affinities: the role of dispersion forces, electrostatics, and orbital interactions. *Phys. Chem. Chem. Phys.* **8**, 1357–1370 (2006).
39. Samanta, J. et al. An ultra-dynamic anion-cluster-based organic framework. *Chem* **8**, 253–267 (2022).
40. Chen, C. X. et al. Dynamic spacer installation for multirole metal–organic frameworks: a new direction toward multifunctional MOFs achieving ultrahigh methane storage working capacity. *J. Am. Chem. Soc.* **139**, 6034–6037 (2017).
41. Zhai, Q. G., Bu, X., Zhao, X., Li, D. S. & Feng, P. Pore space partition in metal–organic frameworks. *Acc. Chem. Res.* **50**, 407–417 (2017).

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026

Methods

High-throughput crystallization

High-throughput crystallization was carried out using solvothermal and vapour diffusion methods. The linkers and acid were first prepared as stock solutions. Different sample conditions with varying equivalents of HCl and solvents were then established using an OT-2 liquid-handling robot. After crystallization, the physical state of each sample was evaluated using an optical microscope. For crystalline solids, unit cell parameters and space groups were then compared across samples to determine whether the crystals corresponded to the same structure. The experimental outcomes were categorized as (1) clear solution: no solid observed in the vial; (2) amorphous or microcrystal: precipitate formed, but no single crystals suitable for mounting; (3) poor-diffracting crystal: crystals that have diffraction spots but were insufficient for space group determination or structure solution; and (4) well-diffracting crystal: crystals that have good diffraction for structure determination.

Solvothermal crystallization. A stock solution of the amine (100 μ l, 20 mM in dimethylformamide (DMF)) was first transferred into 4-ml glass vials. Then, 20, 60, 100 and 200 μ l HCl stock solution (100 mM in MeOH) were added to different vials, corresponding to 1–10 equiv. HCl to linker. To ensure a constant volume of methanol for all samples, 180, 140, 100 or 0 μ l MeOH was added to the respective vials. Afterwards, different antisolvents (800 μ l) were introduced. The vials were sealed and heated in an oven at 80 °C for 10 h and then allowed to cool slowly to room temperature.

Vapour diffusion crystallization. A stock solution of the amine (100 μ l, 50 mM in DMF) was first mixed with 500 μ l HCl stock solutions (10, 30, 50 and 100 mM in MeOH). Then, the mixed solutions (60 μ l) were transferred into 1-ml glass insert tubes, followed by the addition of 100 μ l MeOH to each tube. The insert tubes were then placed inside larger vials containing the corresponding antisolvent and sealed. The vials were left at room temperature for 1 week to allow the antisolvents to gradually diffuse into the inner tubes.

EDA analysis

EDA was performed using the sobEDA method⁴², implemented in the Multiwfn 3.8(dev) software package^{43,44}. All underlying quantum chemical calculations were carried out using the Gaussian 16 (Revision A.03) program package⁴⁵. Calculations were conducted using the Barkla2 High Performance Computing facilities at the University of Liverpool.

All fragments used for the EDA calculations were extracted directly from the crystal structures and the hydrogen atom positions were optimized before energy calculations. The EDA was divided into two parts: the non-linking part and the charged ionic cluster. For the non-linking part, dimer fragments of TAHP or TAHPMe linkers with close contacts were selected and the charged NH_3^+ groups were omitted to eliminate the influence of electrostatic charges on the non-linking interactions. For the ionic clusters (N_4Cl_4 or $\text{N}_{12}\text{Cl}_{12}$), representative fragments were selected and the non-linking portions of the linkers were removed. Hydrogen atoms were added to the carbon position that was bonded to the removed fragments.

sobEDA calculations were subsequently performed on the selected linker dimers and ionic clusters. For the non-linking linker dimers, each monomer was treated as an individual fragment. For the charged ionic clusters, each ammonium unit and chloride ion was treated as an individual fragment; therefore the N_4Cl_4 and $\text{N}_{12}\text{Cl}_{12}$ clusters were described by 8 and 24 fragments, respectively. The energy of each $\text{R}-\text{NH}_3^+\cdots\text{Cl}^-$ ion pair was obtained by dividing the total interaction energy by the number of ion pairs in the system. The interaction between these two fragment systems was evaluated at the B3LYP⁴⁶/6-311+G(2d,p)⁴⁷ level of theory using complex basis functions to suppress basis set superposition error. To accurately account for

non-covalent van der Waals interactions, Grimme's D3 empirical dispersion correction with Becke–Johnson damping (D3BJ) was included^{48,49}. Molecular structures were visualized and all isosurfaces were rendered using the VMD 1.9.3 program⁵⁰.

Data availability

All data supporting the findings of this study are available within the Article and its Supplementary Information and can also be obtained from the corresponding authors upon reasonable request. Crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre (CCDC) under deposition numbers CCDC 2515978 (TAHP-Cl- α), 2515979 (TAHP-Cl- β), 2515980 (TAHPMe-Cl-nc), 2515981 (TAHPMe- N_4Cl_4), 2515982 (TAHPMe- N_{12}Cl_7 -A), 2515983 (TAHPMe- N_{12}Cl_7 -C), 2515984 (Cyclohexane@TAHPMe- $\text{N}_{12}\text{Cl}_{12}$), 2515985 (*n*-Heptane@TAHPMe- $\text{N}_{12}\text{Cl}_{12}$), 2515986 (TPA), 2515987 (TPA- N_6Cl_7) and 2515988 (TPA- N_{12}Cl_9). Copies of the data can be obtained free of charge via <https://www.ccdc.cam.ac.uk/structures/>. Source data are provided with this paper.

References

- Lu, T. & Chen, Q. Simple, efficient, and universal energy decomposition analysis method based on dispersion-corrected density functional theory. *J. Phys. Chem. A* **127**, 7023–7035 (2023).
- Lu, T. & Chen, F. Multiwfn: a multifunctional wavefunction analyzer. *J. Comput. Chem.* **33**, 580–592 (2012).
- Lu, T. A comprehensive electron wavefunction analysis toolbox for chemists, Multiwfn. *J. Chem. Phys.* **161**, 082503 (2024).
- Frisch, M. J. et al. Gaussian 16, Revision A.03 (Gaussian, Inc, 2016).
- Stephens, P. J., Devlin, F. J., Chabalowski, C. F. & Frisch, M. J. Ab initio calculation of vibrational absorption and circular dichroism spectra using density functional force fields. *J. Phys. Chem.* **98**, 11623–11627 (1994).
- Frisch, M. J., Pople, J. A. & Binkley, J. S. Self-consistent molecular orbital methods 25. Supplementary functions for Gaussian basis sets. *J. Chem. Phys.* **80**, 3265–3269 (1984).
- Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J. Chem. Phys.* **132**, 154104 (2010).
- Grimme, S., Ehrlich, S. & Goerigk, L. Effect of the damping function in dispersion corrected density functional theory. *J. Comput. Chem.* **32**, 1456–1465 (2011).
- Humphrey, W., Dalke, A. & Schulten, K. VMD: visual molecular dynamics. *J. Mol. Graph.* **14**, 33–38 (1996).

Acknowledgements

We acknowledge the assistance provided by Research IT and the use of the Barkla High Performance Computing facilities at the University of Liverpool. We thank the Diamond Light Source for beamline allocation I19 (CY36629) for crystal screening.

Author contributions

A.I.C. conceived the initial idea and supervised the project. Y.-L.L. led the experimental work and characterization of the materials. Y.C. performed the EDA calculations. Y.-L.L. and H.Q. conducted the single-crystal X-ray diffraction analysis and solved the structures. Y.-L.L. and L.Z. synthesized TAHPMe and TPA. Y.-L.L., Y.C., H.Q., L.Z., M.O'S., T.F. and A.I.C. analysed the data and prepared the paper. All authors discussed the results and contributed to the paper.

Funding

This project received funding from the Engineering and Physical Sciences Research Council (grant EP/V026887/1). A.I.C. thanks the Royal Society for a Research Professorship (RSRP\S2\232003).

Competing interests

The authors declare no competing interests.

Additional information

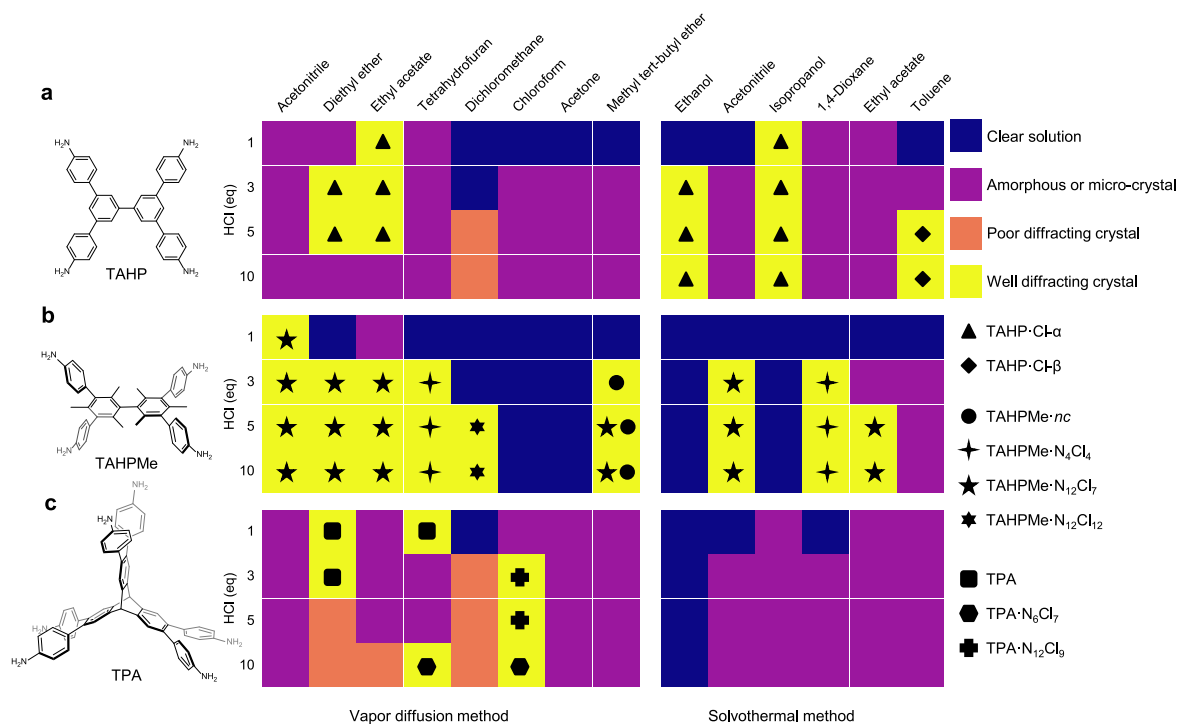
Extended data is available for this paper at <https://doi.org/10.1038/s41557-026-02248-w>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41557-026-02248-w>.

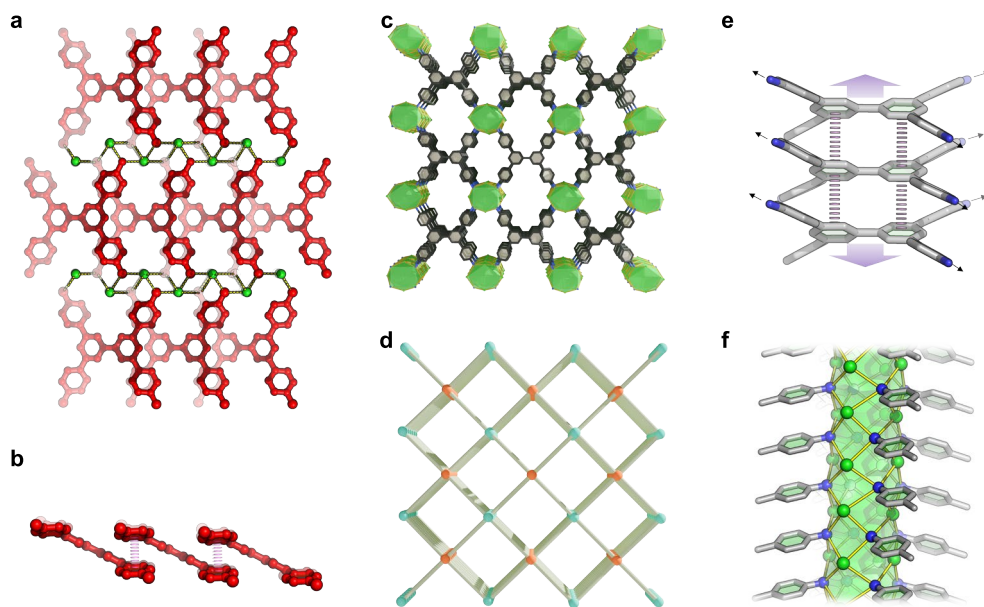
Correspondence and requests for materials should be addressed to Thomas Fellowes or Andrew I. Cooper.

Peer review information *Nature Chemistry* thanks Vincent Guillerm and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

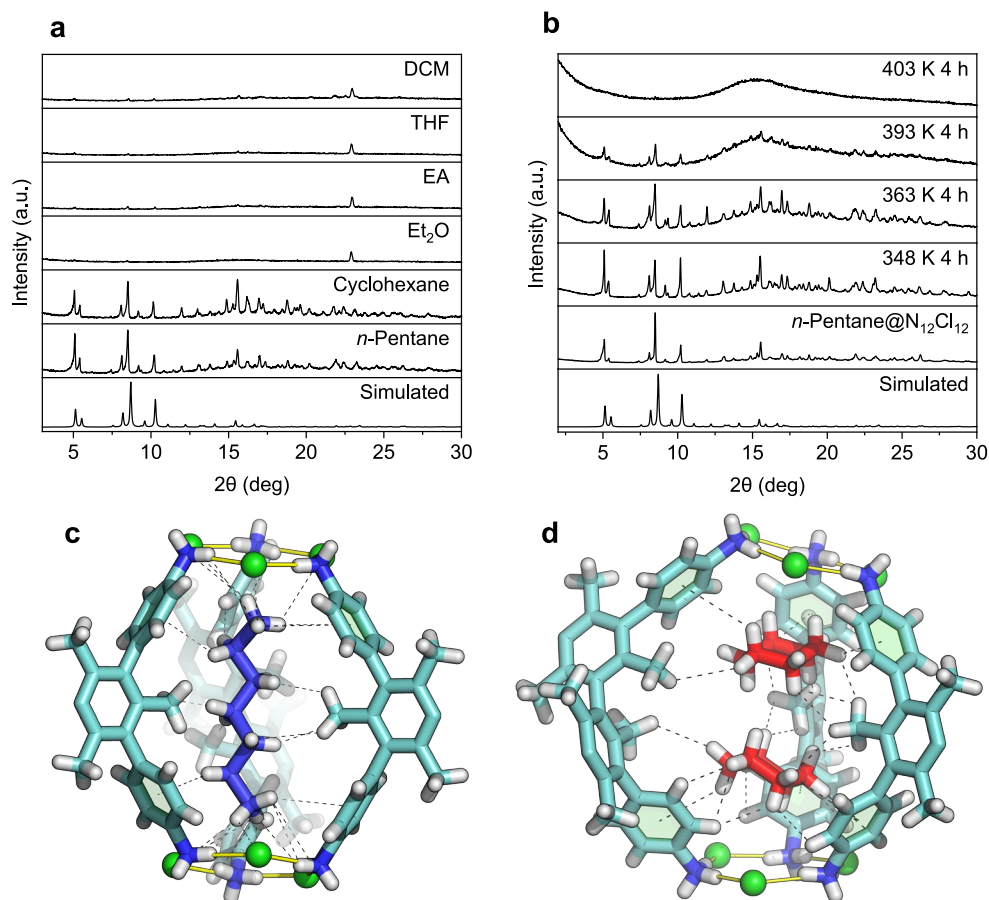


Extended Data Fig. 1 | High throughput crystallization. a, TAHP. b, TAHPMe. c, TPA. Substance states and obtained structures are denoted with different colour and symbols respectively.



Extended Data Fig. 2 | Non-metal organic frameworks assembled from planar 4-c TAHP. **a**, The packing of TAHP·Cl- α (only one disordered part is shown for clarity). **b**, Head-to-tail packing mode in TAHP·Cl- α . **c**, The packing of TAHP·Cl- β . **d**, **sql-b** topology of TAHP·Cl- β . **e**, Face-to-face stacking mode in TAHP·Cl- β . **f**, Tubular structures formed by anilinium chloride in TAHP·Cl- β . Green and blue

spheres represent Cl and N atoms, respectively. Purple dashed lines indicate the non-linking π - π stacking interactions between linkers. Yellow sticks indicate electrostatic and hydrogen-bonding interactions between N and Cl centres ($\text{N}\cdots\text{Cl}$ distances < 3.4 Å).



Extended Data Fig. 3 | Guest-stabilized TAHPMe-N₁₂Cl₁₂. **a**, PXRD patterns of TAHPMe-N₁₂Cl₁₂ samples exchanged with different solvents and subsequently dried under vacuum for 1 h. **b**, Thermal stability of n-pentane@TAHPMe-N₁₂Cl₁₂ after heating at various temperatures for 4 h. **c**, n-heptane@Cavity-1 in

TAHPMe-N₁₂Cl₁₂. **d**, 2-cyclohexane@Cavity-1 in TAHPMe-N₁₂Cl₁₂. Light blue stick denotes TAHPMe linkers; green and blue spheres represent Cl and Na atoms, respectively. The encapsulated n-heptane and cyclohexane are shown as blue and red sticks, respectively.